

Circular maximum Subarray :-

$i=0, \text{sum} = 0, \text{minSum} = 0$
 $i=1, \text{sum} = 0, \text{minSum} = 0$

$i=2, \text{sum} = -1, \text{minSum} = -1$

$i=3, \text{sum} = -3, \text{minSum} = -3$

$i=4, \text{sum} = -1, \text{minSum} = -3$

$i=5, \text{sum} = -9, \text{minSum} = -9$

$i=6, \text{sum} = -3, \text{minSum} = -9$

$i=7, \text{sum} = 0, \text{minSum} = -9$

$i=8, \text{sum} = -100, \text{minSum} = -100$

① (2, 3, 4)

$1+2=3$

②

(6, 1, 2, 4)

→ (Min^m subarray sum)
 (Min^m sum of array) $\rightarrow [1, 2, 3, 4]$

(5, -1, 6, -2)
 (3, 5, -1, -2, 2, -8, 6, 10, -100)
 (Sum) {min} Max. min (with, an)

for
for

minSum = INT_MAX; sum = 0;

for (i = 0 → n) {

sum = sum + num[i];

minSum = min(minSum, sum);

if (sum > 0) sum = 0;

}

if (sum > 0)
sum = 0;

∅

sum = ∅

0

1 2 3 ~

12/11/21

Ques. Circular Subarray Sum :-

$[8, -8, 9, -9, 10]$

max :-

max sum

sum

min :-

min sum

sum

$[8, -8, 9, -9, 10]$

$10 - 9$
 $S_1 + S_2 = S$
 $18 + (-9) = 10$
 $9 = 10$

$S_2 = -8$

8

0

-9

0

-9

19

8

0

-9

0

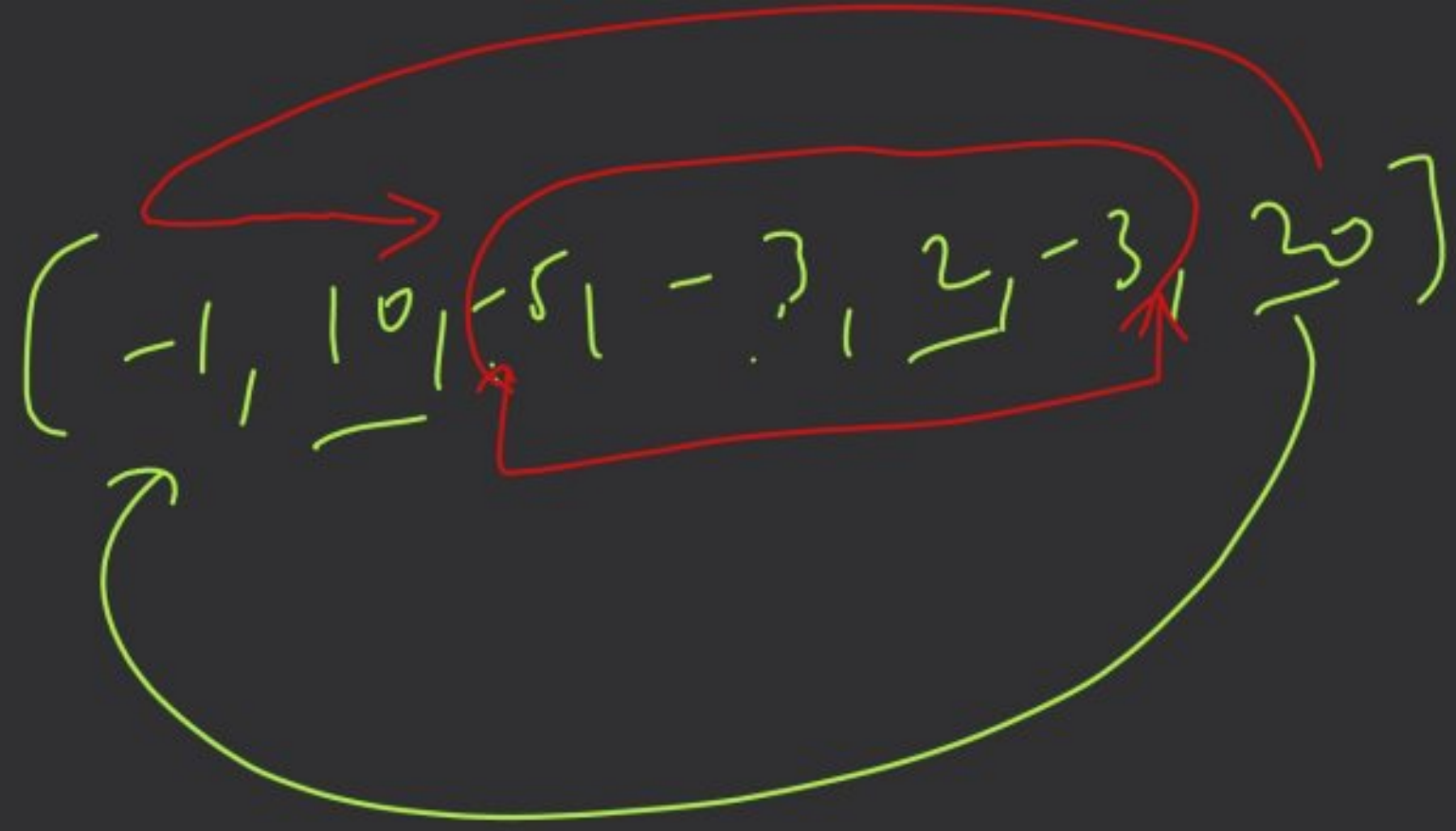
9

$\rightarrow$ min -9
 Take 10

18

10

$19 - 9 = 10$



$S1 \rightarrow \text{max sum subarray}$

$S2 \rightarrow \text{min subarray sum}$

$$S1 + S2 = S \quad (\text{total sum}) \quad \left[\text{circular only} \right]$$

$26 \ 19 \ 29 \ 24 \ 21 \ 23 \ 20$

$$\uparrow S1 = S - (S2) \downarrow$$

(Total sum = 20)

$$S1 = S - S2$$

$$S1 = S - \underline{S2}$$

$\downarrow$
Word

$$\uparrow n = 9 - (5) \downarrow$$

[Good Bad Good]

$$[5, -2, 3, 4]$$

$$[\text{Good} \quad \text{Bad} \quad \text{Good}]$$

$$10 - (-2)$$

$$12$$

$$\text{Circular max sum} = \text{Total} - \text{Min subarray sum}$$

$$[-1, -3, 10, 5, 4, 6, -10, 100]$$

$$204$$

$$204 - (-20) = 224$$

$$-1 \quad (-1, -3, -1)$$

$$-9$$

$$19 - 5$$

Product of zero except self →

Input: [1 2 3 4]

Output: [24 12 8 6]

[4 2 6 3]

→ [36 72 24 48]

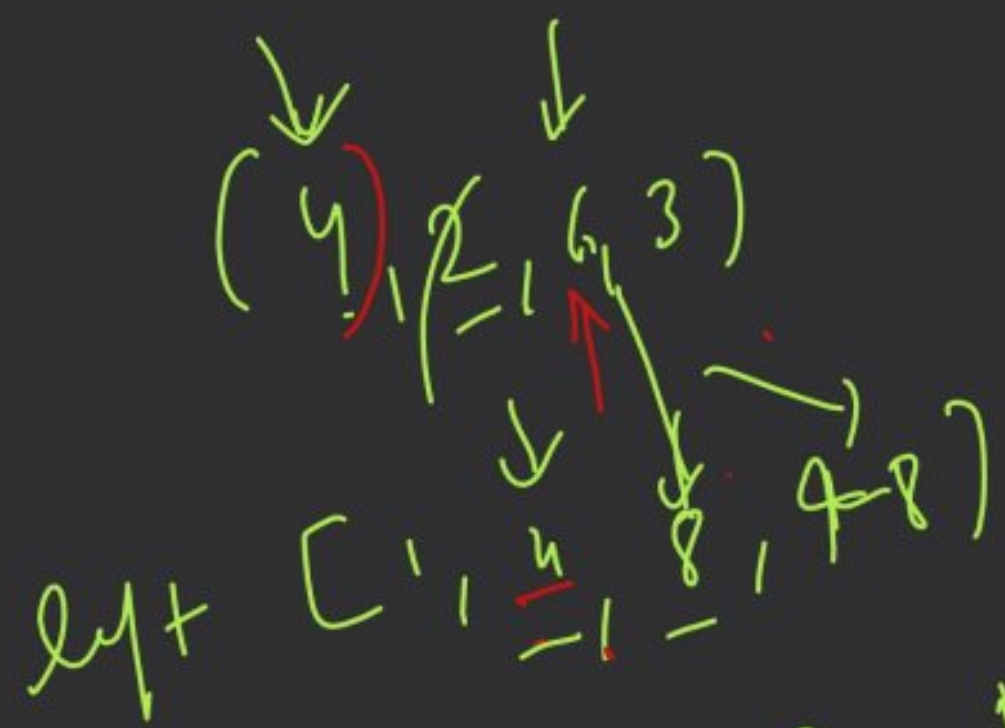
$$0 \rightarrow i-1) * (i+1) * (n-1)$$

$$i=0, 1 \times 2 \times 6 \times 3$$

$$i=1, 4 \times 6 \times 3$$

$$i=2, 4 \times 2 \times 3$$

$$i=3, 4 \times 2 \times 6 \times 1$$



left $[1, 4, 8, 4, 8]$
right $[36, 18, 3, 1]$

$$\text{ans}[i] = \text{left}[0 \dots i-1] * \text{right}[i+1 \dots n-1]$$

$$\text{ans}[2] = 8 \times 3 = 24$$

$[7, 2, 5, 6]$
 $left = [1, 7, 7 \times 2, 7 \times 2 \times 5]$
 $right = [2 \times 5 \times 6, 5 \times 6, 6, 1]$

$ans = [60, 210, 84, 70]$

```

left[0] = 1
for (int i = 1; i < n; i++) {
    left[i] = left[i-1] * nums[i];
}

```

$ans[0] = \text{nums}(1) \times \text{nums}(2) \times \text{nums}(3) = 2 \times 5 \times 6$
 $ans[1] = \text{nums}(0) \times \text{nums}(2) \times \text{nums}(3) = 7 \times 5 \times 6$
 $ans[2] = \text{nums}(0) \times \text{nums}(1) \times \text{nums}(3) = 7 \times 2 \times 6$
 $ans[3] = \text{nums}(0) \times \text{nums}(1) \times \text{nums}(2) = 7 \times 2 \times 5$

$ans[i] = left[i] * right[i]$

```

right[n-1] = 1;
for (i = n-2; i >= 0; i--) {
    right[i] = right[i+1] * nums[i+1];
}

```

$n = \text{num.size};$
 $int left = new int(n);$
 $int right = new int(n);$

//ans array

```

for (int i = 0; i < n; i++) {
    ans[i] = left[i] * right[i];
}

```

$O(n) + O(n) + O(n)$
 $3 \times O(n) \approx O(n)$

Maximum Sum Circular Subarray

C++ to Java Conversion

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Sum Grid Partition I - Lee

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Code

Java

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```
1 class Solution {
2     public int maxSubarraySumCircular(int[] nums) {
3         int total = 0;
4
5         // int maxSum = Integer.MIN_VALUE, currMax = 0;
6         int minSum = Integer.MAX_VALUE, currMin = 0;
7
8         for (int num : nums) {
9
10            // Min Kadane
11            currMin += num;
12            minSum = Math.min(minSum, currMin);
13            if (currMin > 0) currMin = 0;
14
15            total += num;
16        }
17
18        int ans = Math.max(maxSum, total - minSum);
19        return ans;
20    }
21 }
```

Handwritten notes and diagrams:

- A large green circle contains the array $[-3, 100, 100, 20]$.
- An arrow points from the second 100 to a smaller circle containing -100 , which is labeled "Bad".
- Below this, the calculation $17 - (-100)$ is written.
- Below that, the calculation $17 + 100 - 100$ is written.
- A green arrow points from the `total += num;` line in the code to the text "(Total sum)".

Help in coding > Saved

Ln 19, Col 20